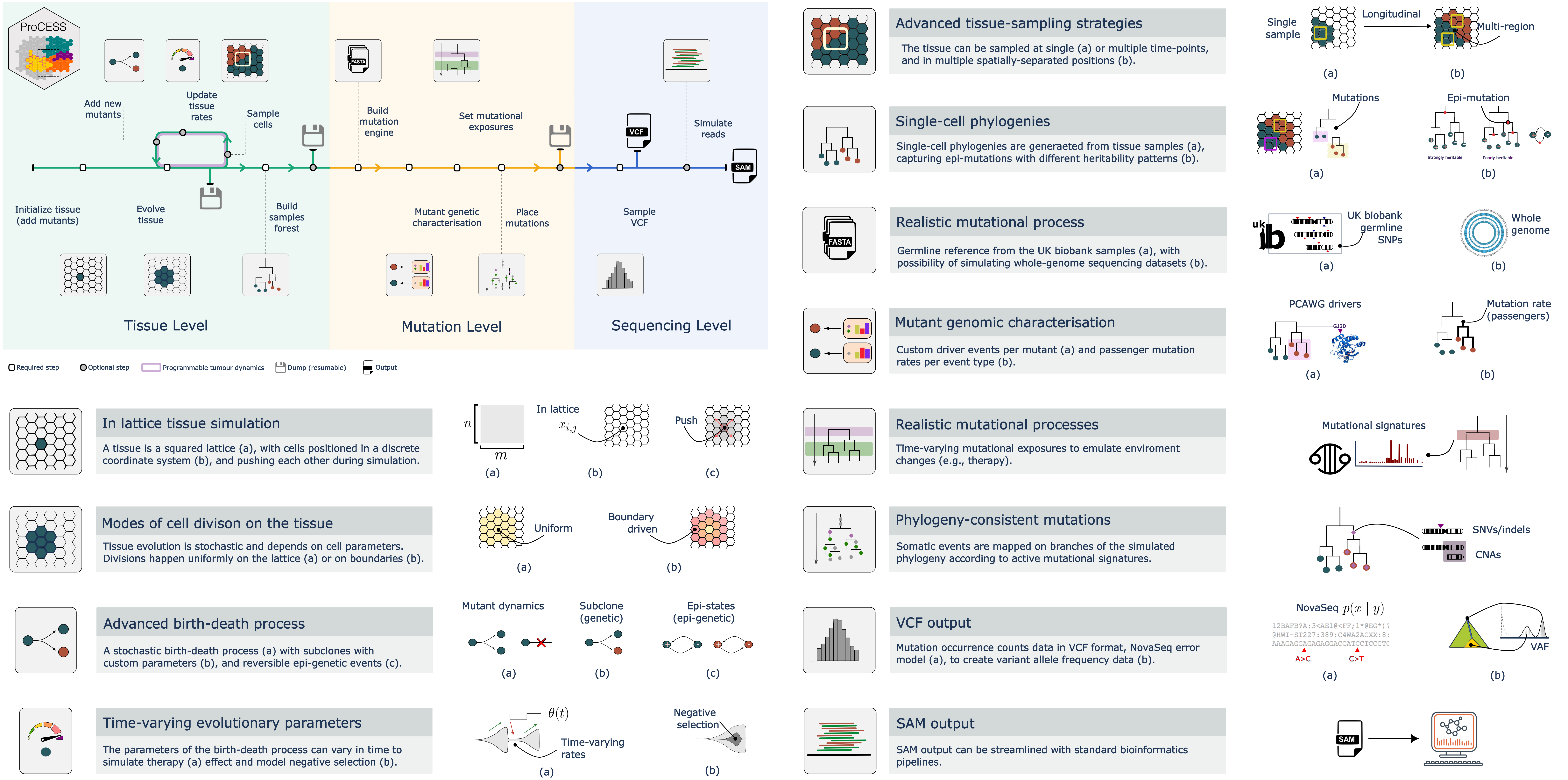
ProCESS
Add new mutants
Update tissue rates
Sample cells
Build mutation engine
Set mutational exposures
Simulate reads
VCF
SAM
FASTA
Initialize tissue (add mutants)
Evolve tissue
Build samples forest
Mutant genetic characterisation
Place mutations
Sample VCF
Tissue Level
Mutation Level
Sequencing Level
Required step
Optional step
Programmable tumour dynamics
Dump (resumable)
Output
In lattice tissue simulation
A tissue is a squared lattice (a), with cells positioned in a discrete coordinate system (b), and pushing each other during simulation.
n
m
In lattice
$x_{i,j}$
Push
(a)
(b)
(c)
Modes of cell divison on the tissue
Tissue evolution is stochastic and depends on cell parameters. Divisions happen uniformly on the lattice (a) or on boundaries (b).
Uniform
Boundary driven
(a)
(b)
Advanced birth-death process
A stochastic birth-death process (a) with subclones with custom parameters (b), and reversible epi-genetic events (c).
Mutant dynamics
Subclone (genetic)
Epi-states (epi-genetic)
(a)
(b)
(c)
Time-varying evolutionary parameters
The parameters of the birth-death process can vary in time to simulate therapy (a) effect and model negative selection (b).
$\theta(t)$
Time-varying rates
Negative selection
(a)
(b)
Advanced tissue-sampling strategies
The tissue can be sampled at single (a) or multiple time-points, and in multiple spatially-separated positions (b).
Single sample
Longitudinal
Multi-region
(a)
(b)
Single-cell phylogenies
Single-cell phylogenies are generaeted from tissue samples (a), capturing epi-mutations with different heritability patterns (b).
Mutations
Epi-mutation
Strongly heritable
Poorly heritable
(a)
(b)
Realistic mutational process
Germline reference from the UK biobank samples (a), with possibility of simulating whole-genome sequencing datasets (b).
uk biobank
UK biobank germline SNPs
Whole genome
(a)
(b)
Mutant genomic characterisation
Custom driver events per mutant (a) and passenger mutation rates per event type (b).
PCAWG drivers
G12D
Mutation rate (passengers)
(a)
(b)
Realistic mutational processes
Time-varying mutational exposures to emulate enviroment changes (e.g., therapy).
Mutational signatures
Phylogeny-consistent mutations
Somatic events are mapped on branches of the simulated phylogeny according to active mutational signatures.
SNVs/indels
CNAs
VCF output
Mutation occurrence counts data in VCF format, NovaSeq error model (a), to create variant allele frequency data (b).
NovaSeq $p(x \mid y)$
A>C
C>T
VAF
(a)
(b)
SAM output
SAM output can be streamlined with standard bioinformatics pipelines.
SAM